

GEOLOGIC HAZARD

Soil Liquefaction Zones

During strong shaking, loose water-saturated soil can temporarily lose strength and behave more like a fluid.



1 Understand the factor

What it is - and why it matters

Liquefaction occurs when strong shaking raises water pressure in loose, saturated, usually sandy or silty soil. The soil may lose bearing strength, settle unevenly, spread laterally toward a river or bay, or eject sand and water at the surface. A liquefaction zone is a screening area, not a guarantee that every parcel will fail. Risk depends on soil layers, groundwater depth, shaking intensity, foundation type, and ground improvement.

How risk can reach a household

- Loss of bearing strength can cause a foundation to sink, tilt, rotate, or separate from utility connections.
- Lateral spreading can pull land apart near shorelines, channels, levees, embankments, and steep grade changes.
- Buried pipes, tanks, manholes, and utility vaults can break, float, or become misaligned.
- Settlement can damage roads, sidewalks, drainage systems, and neighboring parcels even when the house remains standing.

What people often miss

- Liquefaction can occur far from the earthquake fault wherever strong shaking and susceptible soil coincide.
- Reclaimed land, old stream channels, bay fill, young alluvium, and high groundwater are common indicators.
- Groundwater levels vary by season, drought, pumping, irrigation, and nearby construction or recharge projects.
- A general hazard-zone map cannot replace a geotechnical investigation for a specific lot or planned addition.

? Five checks before buying

1

Review state or local liquefaction maps and determine whether the parcel, access road, and utilities are inside the zone.

2

Obtain available geotechnical reports, boring logs, foundation plans, and settlement or underpinning records.

3

Look for sloping floors, patched foundation cracks, separated utility lines, recurring drainage issues, and tilted exterior features.

4

Ask a geotechnical or structural professional whether deep foundations, densification, drains, or other mitigation were used.

5

Confirm how earthquake coverage treats settlement, ground failure, utilities, and additional living expense.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 02

H Health impacts

- Primary health danger comes from structural movement, falling objects, fire, broken glass, and damaged roads during an earthquake.
- Gas leaks and electrical failures can create fire, explosion, carbon monoxide, and smoke hazards.
- Broken water and sewer lines can cause loss of sanitation, contaminated standing water, and delayed emergency care.
- Extended repair disputes or displacement can contribute to stress, sleep disruption, and financial hardship.

P Property impacts

- Differential settlement may distort the foundation, walls, doors, windows, floors, plumbing, and drainage systems.
- Neighborhood streets, levees, and lifelines may fail, reducing access and habitability even if the house has limited damage.
- Repair can require underpinning, piles, soil improvement, demolition, utility reconstruction, or elevation work.
- Prior settlement and geotechnical findings may affect disclosures, appraisal, financing, renovation feasibility, and resale.

I Insurance implications

- Liquefaction is generally treated as earthquake-related earth movement, which standard homeowners insurance usually excludes.
- Soil stabilization performed before a loss is normally a maintenance or capital cost, not an insured repair.
- Earthquake policies may cover direct physical loss but can have large deductibles and limitations for land, foundations, and exterior systems.
- Insurers and lenders may request engineering information when there is known settlement, prior claims, or unusual foundations.

Possible long-term health implications of buying nearby

- Proximity alone does not create a daily health exposure; the main concern is severe ground failure during a future earthquake.
- Uneven floors and recurring movement can create fall hazards and chronic worry, particularly for older adults or people with mobility limitations.
- If an event damages sewers, water lines, or drainage, residents can face mold, contaminated water, dust, and sanitation problems during recovery.
- A sound geotechnical design can materially reduce risk, so investigate the actual soil and foundation rather than relying only on a map label.

Evidence lens

The mechanism of liquefaction is well established. Whether a particular parcel will liquefy, and how severely, requires site-specific soil, groundwater, foundation, and shaking information.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 02

V Verify the actual risk

- Obtain the mapped-zone report and any site-specific geotechnical studies; verify the date and assumptions.
- Review drainage, groundwater, foundation, settlement, and repair records for the house and nearby infrastructure.

P Protect the property

- Use qualified engineers for additions, retaining walls, pools, and major remodels that could change loads or drainage.
- Maintain positive drainage, flexible utility connections, and accessible shutoffs; repair leaks that can saturate soil.

M Monitor and respond

- Track new cracks, floor-level changes, sticking doors, pipe separations, yard depressions, and recurring wet spots.
- After strong shaking, avoid damaged roads and inspect gas, water, sewer, foundation, and electrical systems.

I Prepare insurance records

- Compare earthquake coverage for dwelling, foundation, contents, code upgrades, utilities, and temporary housing.
- Keep engineering reports and baseline photographs to distinguish pre-existing conditions from event damage.

REAL-LIFE EXAMPLE

1989 Loma Prieta Earthquake: Marina District

October 17, 1989

During the magnitude 6.9 Loma Prieta earthquake, liquefaction and ground deformation severely damaged parts of San Francisco's Marina District, which included artificial fill and water-saturated soils. Buildings settled or tilted, utilities failed, and fire complicated emergency response.

Why it matters: The case demonstrates that soil conditions can concentrate damage far from the earthquake's epicenter and that utility resilience and foundation design are central to recovery.

[Open official incident record - usgs.gov >](#)**Questions to resolve before making a decision**

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3. Look for sloping floors, patched foundation cracks, separated utility lines, recurring drainage issues, and tilted exterior features.
4. Ask a geotechnical or structural professional whether deep foundations, densification, drains, or other mitigation were used.

Official sources and mapping tools[USGS liquefaction explanation - usgs.gov >](#)[USGS Marina District image and context - usgs.gov >](#)[California seismic hazard mapping - conservation.ca.gov >](#)